

From Falsification to a Validated Apparatus: Mean Reversion in Commodity Calendar Spreads Inside Structurally Trendy Markets

Adaptive Mean-Reversion (AMR) Research Programme
Internal Research Dossier — Compiled Synthesis

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Abstract

We report the trajectory of a falsification-first research programme over the interval in which its flagship regime-transition object was killed and a deployment-relevant mean-reversion edge was, for the first time, validly confirmed. The programme’s original object—a pre-reversion “stabilization” morphology in the equilibrium residual—is falsified across twelve instruments spanning five habitats and three sampling resolutions, and the falsification is extended to a sixty-two file cross-habitat sweep with zero exceptions; a single mechanism, *selection-on-deviation*, reproduces the entire result with no transition object required. A provenance audit then disqualifies every pre-existing spread in the data set, and a controlled study shows that even a strictly causal, pre-registered *rolling* hedge-ratio estimator *manufactures* the variance-ratio structure it purports to measure—hedge-ratio update noise accounts for 82–97% of the differenced-spread variance. Restricting construction to definitional unit-ratio calendar spreads removes this artifact and furnishes a real-data positive control. On a natural-gas front calendar the apparatus confirms genuine short-horizon stochastic mean reversion ($VR(20) = 0.448$, $p = 0.005$ against random-walk, GARCH and moving-average microstructure nulls), corroborated on an independently constructed Brent calendar. A rolling-local analysis finds the natural-gas signal statistically persistent across nineteen years (pooled standardized distance -0.627 , $t = -4.37$, $p \approx 2 \times 10^{-4}$) and economically *unecomic*: a causal mean-reversion book is break-even before costs and negative after. We maintain throughout a strict separation of three claims—that the *apparatus works*, that the *signal is real*, and that the edge *clears costs*—and conclude that the first two are earned and the third is not. We close with the corrected methodological framework, the open problems, and three forward hypotheses.

Keywords: mean reversion; variance ratio; commodity calendar spreads; theory of storage; surrogate-relative inference; look-ahead bias; falsification.

Status legend. [FROZEN] decided, not reopened without justification; [KILLED/ARCHIVED] falsified-in-form; [CONDITIONAL SURVIVAL] valid in named regimes with a stated kill trigger; [PERSISTENT-BUT-UNECONOMIC] statistically real, does not clear costs; [OPEN] genuinely unresolved; [DEFERRED] deprioritized with a named reopen trigger. Confidence labels (trustworthiness of evidence): LOW, LOW-MEDIUM, MEDIUM, MEDIUM-HIGH, HIGH.

1. Introduction

The programme seeks a narrow but genuine edge: to trade mean reversion inside markets that look structurally trendy, and to *stop*—treating stopping as a successful outcome—if the edge is not there. This dossier covers the interval bounded by two events. It opens at the falsification of the programme’s flagship regime-shift object and closes at the present research frontier.

The narrative is a single arc: *diagnosis, falsification, reconstruction, corrected framework, and forward path*.

The organising discipline of the report is a separation that the field routinely blurs. A research result can be a success in three different and non-interchangeable senses:

Research win. The measurement apparatus works: it has power, selectivity, a calibrated false-positive rate, and the ability to discriminate genuine mean reversion from microstructure noise, demonstrated on controls of known construction.

Empirical win. The signal is real: a specific instrument separates from a matched surrogate in the mean-reverting direction, robustly across regimes.

Deployment win. The edge clears costs: a causal trading rule has positive net expectancy after all-in transaction costs over a forward hold-out.

As of this writing the programme holds a research win and an empirical win. It does *not* hold a deployment win. Every claim below is filed under exactly one of these headings.

The principal methodological commitments are inherited and unrelaxed: at time t only information available at t may enter a causal computation; every statistic is read *surrogate-relative* (against a matched null run through identical extraction), never against a fixed constant or the instrument’s own history; hypotheses, windows, and decision rules are fixed before data are touched; and a falsified idea is named and closed rather than silently revived.

2. The Falsification of the Pre-Reversion Morphology

2.1 The object and its pre-registered signature

The original Layer-2 object posited that an extended deviation from a latent equilibrium price μ^* *loses momentum and stabilizes before reverting*, leaving a measurable pre-reversion morphology. This was operationalized—deliberately as an observational existence question, not a detector—on the causal one-step innovation stream of an anchored Kalman equilibrium filter. The signature was fixed in advance: in high- $|z|$ causal pre-windows (selected by a symmetric partition on the causal standardized deviation, $\theta \in \{1.0, 1.5, 2.0\}$, window $W = 30$), relative to a matched null, all three of the following would hold:

- (D1) innovation variance $\downarrow$ (residual stabilization);
- (D2) innovation autocorrelation $\text{ACF}(1) \downarrow$ (deviation losing momentum);
- (D3) directional efficiency $\downarrow$ (price chops rather than continues).

A “signature” required all three negative. The claim was distributional—about an ensemble, never a single bar—so that its meaning did not depend on *when* it was read.

2.2 The method was validated before it was trusted

Synthetic calibration established that the measurement was not blind: it produced no false positive on ordinary Ornstein–Uhlenbeck (OU) or random-walk mean reversion (all standardized mean differences $|d| \leq 0.15$), and it fired loudly and in the correct direction on a clean stabilize-then-revert positive control ($d \approx -1.9$ for innovation variance, -2.1 to -2.5 for $\text{ACF}(1)$, -0.9 for directional efficiency). A first, mis-built positive control was rejected and recorded rather than quietly discarded. The method’s confidence entered the real-data phase at MEDIUM–HIGH; the existence of the object itself remained UNTESTED.

2.3 The kill

The decisive instrument was committed in advance: a canonical mean-reverting equity pair spread. The committed rule stated that directional continuation in that instrument, like the rest of the cohort, would meet the kill condition. It did. Across twelve instruments spanning five habitats, two to three resolutions, two comparison methods and three thresholds, directional efficiency was *positive in every one*—high-deviation windows were *more* directionally efficient, the exact opposite of the pre-registered stabilization. The within-instrument, confound-controlled comparison reproduced the sign, ruling out a null-calibration artifact.

A single null reproduces the entire result with no transition object: *selection-on-deviation*. Conditioning on $|z| \geq \theta$ samples paths that arrived at an extreme *by directional travel*; those pre-windows are directionally efficient by construction. The effect is the Lo–MacKinlay [4] cross-autocovariance selection trap, and continuation around extremes is the expected result at these horizons in the momentum literature [5, 6]. The result is textbook, not anomalous.

A subsequent sweep over sixty-two files and twenty-plus instruments—equities, commodities, foreign exchange and rates at three resolutions—returned sixty-two rejections, zero confirmations, zero inconclusive and zero noise, with the relevant descriptors universally and substantially wrong-signed.

The pre-reversion stabilization morphology is [KILLED/ARCHIVED] (**structural; confidence high in the kill**). Confidence that it reflects a real phenomenon in any tested habitat: $< 5\%$. The programme’s question inverts from *when* reversion ignites to *where* and *whether* reversion robustly exists, and its bottleneck moves from theory to data trustworthiness. The surviving load-bearing rule: residual reversion and residual half-life are *smoother-manufactured*—“did price come back?” is a contaminated question; the valid discriminator is equilibrium stability (did μ^* stay put?), and the return-space variance ratio is the one core statistic that escapes the trap. No transition-morphology variant reopens without a new mechanism and new evidence.

3. The Provenance and Construction Crisis

3.1 Provenance is upstream of every statistic

The kill exposed an inverted ordering: the programme had pursued *when* before establishing *where*, on data whose causality was unverified. A provenance audit, with disposition rules fixed before inspection, returned a binding fact: every deployment-domain spread in the data set was a pre-computed single price column with its constituent legs discarded. With the legs gone, the hedge-ratio’s causality is unrecoverable, and a full-sample hedge ratio—which renders a spread stationary by hindsight construction—is mechanically indistinguishable from genuine mean reversion. Consequently *none* of the eight pre-existing spreads survived to the analysis whitelist; all were contaminated by default. Contamination is the default; trust is earned by a constructive causal proof.

The resolution was a deep cohort of forty-six trusted raw legs across energy, metals, agriculture, foreign exchange and equities, from which seven spreads are constructible under a causal protocol. For the first time the programme held instruments whose causality could be proven by construction rather than assumed.

3.2 The construction law

The audit was generalised into a construction-legality standard governing data only, upstream of every reversion statistic. Its governing question, applied to every derived value at every bar, is whether that value could have existed using only information available at time t . The standard fixes a hedge-ratio ontology ordered from safest to most assumption-laden—unit ratio (definitional, zero look-ahead) $\succ$ structural/economic ratio $\succ$ causal rolling regression $\succ$ causal rolling cointegration $\succ$ dynamic state-space ratio (deferred)—and forbids a full-sample ratio of any kind as evidence. It requires trailing-window estimation lagged to $t - 1$, timestamp (not index) synchronisation, inner-join alignment, and a level-difference representation for any spread that can cross zero (logarithmic and ratio statistics are undefined through zero). It treats naive spread high/low constructed from separately-timed leg extrema as counterfactual states that never existed.

3.3 A causal estimator that manufactures its own answer

The first verdict-capable study built seven spreads under this protocol using a causal, lagged *rolling* ordinary-least-squares hedge ratio on a sixty-bar window, and tested the level-difference variance ratio against random-walk, GARCH and OU surrogates. Every spread passed a future-injection causal acceptance test, and the engine was ground-truth validated. The headline result was nonetheless *inconclusive and construction-defective*. Decompose the differenced spread, $S_t = A_t - \beta_t B_t$:

$$\Delta S_t = \underbrace{(\Delta A_t - \beta_{t-1} \Delta B_t)}_{\text{spread innovation}} - \underbrace{(\beta_{t-1} - \beta_{t-2}) B_{t-1}}_{\text{hedge-ratio update noise} \times \text{trending level}}. \quad (1)$$

The second term—the change in the estimated ratio multiplied by the *level* of a trending leg—was measured at 82–97% of $\text{Var}(\Delta S)$ on the failing spreads, with autocorrelation ≈ 0.81 , injecting a smooth persistent component that drives the variance ratio far above one (spurious super-diffusion). On synthetic ground truth where the true spread is OU (true VR < 1), the true-ratio variance ratios [0.97, 0.91, 0.81, 0.65] were inflated by the rolling estimator to [1.81, 3.68, 5.46, 6.23]; ablating the update term in (1) recovered VR < 1 . The estimator manufactures the property it measures. The cohort yielded zero admissible confirmations: two apparent confirmations were a stale-quote artifact (75.6% flat bars) and a volatility-clustering near-miss (separates from a random walk at $p = 0.005$ but not from a GARCH martingale at $p = 0.094$); the remainder were construction-invalid.

No pre-existing spread is admissible [FROZEN]; the construction law is [FROZEN]. **The rolling-OLS-on-levels estimator is killed as inadmissible** for variance-ratio habitat tests on trending legs (confidence HIGH: synthetic ground truth, ablation, two independent variance decompositions, a monotone window-length gradient). The load-bearing finding is methodological, not about the market: a causal time-varying hedge ratio can leak *interpretively*—manufacturing the measured property—without leaking *temporally*. The deployment-domain reversion premise is thus *neither confirmed nor validly damaged*; the bottleneck moves from data to construction. The only artifact-free construction available is the definitional unit ratio, which both fixes the estimator and supplies a real-data positive control.

4. Apparatus Validation: Natural-Gas Storage Mean Reversion

4.1 The positive-control mandate and the unit-ratio restriction

Every positive control to date had been synthetic, so a negative result was formally indistinguishable from a too-blunt test. The standing requirement is therefore that, before any further negative is credible, the apparatus must *confirm* a known, literature- and economically-anchored real edge. The chosen edge is storage/term-structure mean reversion in a commodity calendar spread—grounded in the theory of storage [1] and the commodity term-premium literature [2]—and the primary instrument is a natural-gas front calendar, $S_t = G_{m1} - G_{m2}$. Construction is restricted to definitional unit-ratio calendar spreads: the two legs are the same commodity, so the one-to-one relationship is definitional rather than estimated, the construction carries zero hedge-ratio degrees of freedom, and it is structurally immune to the artifact of (1).

4.2 The synthetic calibration gate

Because the surrogate ensemble was extended with a moving-average (MA(1)) microstructure null, the calibration was re-earned on synthetic ground truth before any real verdict. All four controls passed (Table 1). The decisive control is the third: a pure random walk plus i.i.d. level noise (zero mean reversion) produces a variance ratio below one that the random-walk and GARCH nulls *cannot* absorb—precisely the bid-ask-bounce artifact a calendar spread inherits. Only the MA(1)-in-increments null discriminates genuine reversion from lag-one microstructure.

Control	Required	Result
True seasonal-OU calendar	confirm (power)	confirms
True random-walk calendar	null at $\approx 5\%$	FPR = 3.0% (200 seeds)
Random walk + i.i.d. noise (zero MR)	MA(1) must reject	rejected by MA(1)
No-seam invariance (clean OU)	VR shift ≈ 0	shift = 0.000

Table 1: Synthetic calibration gate. The 3.0% false-positive rate over 200 seeds is below the nominal 5% and tighter than any single surrogate family because the gate is a conjunction.

4.3 Confirmation

Table 2 reports the verdict matrix (real minus matched surrogate; multiplicity-corrected minimum-variance-ratio p). The natural-gas calendar beats all three martingale-class nulls at the fifth-percentile gate, sitting at the surrogate floor. The confirmation survives causal deseasonalization (the raw confirmation is load-bearing; the deseasonalized strengthening is reported but not relied upon), is consistent across the open and close sampling points, and—critically—is corroborated on an *independently constructed* Brent calendar built from raw legs under a controlled roll rule with no vendor back-adjustment, which confirms at *all* horizons. The apparatus is also selective: it correctly nulls a refined-products back-month calendar at the book horizon ($q \leq 20$, $p = 0.537$) and recovers its reversion only at the long horizon ($q \geq 60$, $p = 0.005$), exactly the back-month storage profile theory predicts.

4.4 The residual open channel

An adversarial review returned *holds-with-caveats*. The one artifact the battery does not fully discriminate is vendor back-adjustment: additive back-adjustment at the monthly roll mechanically compresses long-horizon variance, producing a declining variance ratio that is indistin-

Instrument	Role	n	VR(2, 5, 10, 20)	p_{RW}	p_{GARCH}	p_{MA1}	gate
NG calendar (daily)	primary	4969	0.92, 0.83, 0.61, 0.45	0.005	0.025	0.005	confirm
Brent calendar (built, 60 min)	control	22955	0.91, 0.80, 0.71, 0.63	0.005	0.010	0.005	confirm (all q)
RB calendar (daily)	reference	5170	0.996, 0.993, 0.983, 0.979	0.537	—	—	null (expected)
RB long-horizon	—	5170	$q_{40}=0.63, q_{60}=0.49, q_{120}=0.42$	0.005	—	—	reverts $q \geq 60$

Table 2: Verdict matrix. The natural-gas confirmation is carried at the long horizon ($q = 10, 20$); the self-built Brent calendar confirms at all horizons, the opposite of the long-horizon-only compression signature that vendor back-adjustment would produce.

guishable from storage mean reversion under a level-difference statistic. This channel is *partially mitigated* by the construction-controlled Brent corroboration but *not eliminated* for the natural-gas daily series specifically.

[CONDITIONAL SURVIVAL]: the real-data positive control is **passed in the conditional sense**. **Research win (apparatus works)** — **medium-high**: demonstrated power, selectivity, 3.0% false-positive rate, noise-discrimination, and construction-controlled corroboration; the synthetic-only prior state is materially improved. **Empirical win (signal real)** — **medium**: the confirmation is surrogate-relative and robust, but *clean storage mean reversion* versus *partial vendor-construction artifact* is mitigated, not eliminated. **Deployment win** — **not claimed**: a positive control is not a deployment target.

5. Rolling-Local Deployability: Persistent but Uneconomic

5.1 The trader-relevant question and an honest instrument

Rolling/local estimation is the principal look-ahead and degrees-of-freedom surface—it manufactured the super-diffusion of (1)—so the rolling-local analysis was pre-registered under a binding guard: windows fixed in advance, surrogate-relative under identical extraction, multiplicity controlled, and local survivability defined as persistence across *multiple disjoint* windows rather than appearance in one. A first-draft per-window binary flag was falsified pre-registration on synthetic ground truth (underpowered on genuine OU, firing *higher* on bid-ask bounce, blind to back-adjustment) and replaced—before any real rolling estimate was computed—by a pooled continuous statistic. For each window w ,

$$z_w = \frac{\text{VR}(20)_w - \overline{\text{VR}(20)}_w^{\text{RW}}}{\hat{\sigma}(\text{VR}(20)_w^{\text{RW}})}, \quad \text{pooled mean-}z = \frac{1}{W} \sum_{w=1}^W z_w, \quad (2)$$

with the random-walk band computed in-window so the small-sample variance-ratio bias self-cancels. Synthetic anchors were frozen first: a random-walk null at +0.02 (5–95% band $[-0.32, +0.41]$), genuine moderate reversion (OU, $\phi = 0.95$) at -0.33 , and a back-adjustment splice (fraction 0.5) at -0.99 .

5.2 Results

The natural-gas signal is statistically persistent (Table 3): pooled mean- $z = -0.627$ ($t = -4.37$, $p \approx 2 \times 10^{-4}$) over nineteen disjoint years, with fourteen of nineteen years below the random-walk median, present both before and after 2020. The global half-life is 12.9 bars (≈ 2.6 weeks, inside the tradeable band). And the structure is reversion-favoring: the signal *switches off in five years*—all storage-glut / low-scarcity regimes in which the physical restoring force is

weakest, not acute shocks. A uniform back-adjustment artifact would not selectively switch off in glut years, which is the strongest evidence the persistence is genuine storage mean reversion.

Read	mean- z	n	note
NG yearly (primary)	-0.627	19	$t = -4.37$, $p \approx 2 \times 10^{-4}$; 14/19 below RW median
NG post-2020	-0.438	6	below -0.32: recency holds, attenuated
Brent (built, no back-adj.)	-3.54	8	hourly/off-scale: apparatus control, not NG-level evidence
splice frac. 0.25 / 0.5	-0.657 / -1.053	—	NG (-0.627) $\approx$ the quarter-strength splice anchor

Table 3: Rolling-local persistence. The natural-gas pooled distance is statistically indistinguishable from the quarter-strength splice anchor ($p = 0.84$), so genuine reversion versus partial back-adjustment remains MEDIUM confidence.

The economics, however, are decisive. A frozen causal mean-reversion rule (enter at $|z| \geq 1$, exit at $z = 0$ or a twenty-bar stop, round-trip cost 0.003 spread units) yields, over 200 trades, an average *gross* capture of +0.0004 and an average *net* of -0.0026 per trade (hit-rate 0.53). The signal is break-even before costs and negative after; clearing the cost floor would require a gross improvement of roughly seven-fold that no admissible rule-optimization can supply on this instrument.

[PERSISTENT-BUT-UNECONOMIC] (**confidence medium**). **Empirical win — strengthened, qualified:** the natural-gas calendar sub-diffusion is persistent versus a random walk (HIGH: $t = -4.37$) and reversion-favoringly structured (switches off in gluts, not shocks); that it is genuine storage mean reversion versus partial back-adjustment remains MEDIUM ($p = 0.84$ against the quarter-strength splice). **Deployment win — no** (HIGH: the naive single-instrument book is uneconomic). The deployable object is accordingly redefined from an *instrument* to a *cost-aware book*; whether a portfolio of individually sub-cost, leg-netted spread edges can clear costs is the decisive untested question.

6. Corrected Framework and Forward Path

6.1 A rolling-local ontology under a binding guard

The arc above motivates a change of ontology that is directional, not methodological. The market is treated as piecewise and regime-local rather than as one global object; the operative question is *where* and *when* mean reversion exists, not whether it exists universally. Rolling/local estimation becomes a first-class default but only under the binding guard of Section 5—pre-registered windows, surrogate-relative extraction, multiplicity control, and survivability across multiple disjoint windows—because rolling estimation is the single most dangerous degrees-of-freedom surface in the programme. Global claims now require cross-habitat replication plus a mechanism; local conditional claims are the default currency.

6.2 Deployability as a portfolio property, evaluated after costs

The deployable object is a *book*, not an instrument: the unit of success is positive cost-adjusted portfolio expectancy, evaluated after transaction cost, roll, borrow, capacity and sizing. A spread that mean-reverts but is uneconomic after costs is a non-finding—precisely the status of the natural-gas calendar. Transaction costs, capacity and risk are first-class research outputs, not afterthoughts. This prioritization decides *which* questions earn effort; it never relaxes *how rigorously* a survivor is judged.

6.3 Open problems

No cross-habitat daily replication. [OPEN] The natural-gas confirmation is a single daily instrument; the Brent corroboration is at an off-scale hourly resolution. The unit of evidence is survival across independent habitats, not appearance in one.

Back-adjustment not closed. [OPEN] The natural-gas persistence is statistically indistinguishable from a quarter-strength splice; genuine-reversion confidence stays MEDIUM until the seam is splice-tested or the series is rebuilt from raw legs.

No controlled-ratio pair. [DEFERRED] The deployment construction proper—pairs and cross-asset relative value with a regularized hedge ratio—is untested; the rolling estimator was shown to mishandle it, and a regularized ratio must be validated by equilibrium stability rather than residual reversion.

Trend-death detector. [DEFERRED] A trend-death $\rightarrow$ reversion operationalization returned no content (a permissive 18–20% firing rate, zero of four instruments meeting the pre-committed criteria, and anti-thesis continuation in structural growth equities); the thesis is not falsified but is untested in the right instrument domain, and any redesign must target a firing rate below 5%.

Portfolio economics. [OPEN] The decisive door: whether aggregating sub-cost per-trade calendar edges with leg-netting clears costs is the only demonstrated path from “merely true” to “deployable”.

6.4 Three forward hypotheses

The following are forward directions, not findings; each is open or deferred, carries a reopen trigger, and inherits the multiplicity and surrogate-relative discipline above.

H-Residual-Growth ([DEFERRED]). That the expansion/compression dynamics of the equilibrium residual around *realized* reversions carry market-specific, conditional structure that a matched surrogate does not already produce. This is the legitimate *observational* successor to the killed morphology—strictly descriptive, never a detector; any drift toward predicting reversion from residual shape is a resurrection and is forbidden without a new, independent pre-registration.

H-Confluence ([OPEN]). That a deployable selectivity edge arises not from any single signal but from the confluence of multiple independent, surrogate-relative conditions—each individually weak, jointly informative for *where not to trade*. Its danger is multiple-comparison degrees of freedom; it is admissible only with pre-registered cohorts and cross-habitat out-of-sample replication.

H-Econ-Confluence ([DEFERRED], data-gated, long-term only). The economically-conditioned form: that causal, non-price fundamental variables (inventory surprise, positioning, signed order flow) condition the confluence—the etiology-conditioned reversion principle (inventory/flow $\rightarrow$ reverts; information $\rightarrow$ continues). It is the best-grounded mechanism in the programme and is blocked on data acquisition; a price-level conditioning attempt was already killed as a selection-on-regime artifact, so the admissible form is surprise- or flow-based.

6.5 The long-term object

The terminal ambition is a multi-instrument mean-reversion portfolio engine: a book of independently cost-clearing, surrogate-validated, causally constructed spread edges across deployment

habitats, sized and risk-managed so that net-of-cost expectancy is positive over a forward hold-out. The path is now empirically posable for the first time, because the apparatus is validated and one real edge is confirmed; it is also honestly long, requiring the back-adjustment channel to be closed, replication across daily habitats, validation of the controlled-ratio construction, and a demonstration that a book of individually sub-cost edges can net to a cost-clearing aggregate. None of that is done. The posture is to build that book one validated, cost-aware sleeve at a time—and to stop, and call it a success, the moment the evidence says the book does not pay.

7. Conclusion

Over the interval reported, the programme killed its flagship transition object cleanly and at scale, diagnosed the structural error that produced it (asking *when* before establishing *where*, on untrustworthy data), rebuilt its data foundation, discovered and removed a construction artifact in which a causal estimator manufactured its own signal, and—on a corrected, definitional construction—confirmed for the first time a known real mean-reversion edge with a calibrated, selective, noise-discriminating apparatus. That same edge, read honestly against transaction costs, is persistent but uneconomic as a single instrument. The programme therefore holds a research win and an empirical win and has not earned a deployment win. The remaining path runs through portfolio economics and causal conditioning variables, never through a resurrected timing object. Throughout, stopping where the evidence stops is treated as a successful outcome.

A. Definitions and Core Formulae

Level-difference variance ratio. For a level series S_t that may cross zero,

$$\text{VR}(q) = \frac{\text{Var}(S_t - S_{t-q})}{q \text{Var}(S_t - S_{t-1})},$$

with $\text{VR} < 1$ indicating sub-diffusion (the mean-reverting direction), $\text{VR} \approx 1$ a martingale, and $\text{VR} > 1$ super-diffusion. Read surrogate-relative (real minus matched null under identical extraction), never against a fixed constant.

Surrogate ensemble. Random walk, GARCH(1,1), MA(1) microstructure noise, and OU, each at identical length and extraction; the gate is the multiplicity-corrected minimum-variance-ratio p at the fifth percentile.

Half-life. From a causal first-order autoregressive coefficient ϕ , $t_{1/2} = \ln(0.5)/\ln(\phi)$; the tradeable band is $\approx 3\text{--}40$ bars.

Cost hurdle. A rule is deployable only if the expected net per trade, $\mathbb{E}[\text{gross capture} - \text{round-trip cost}]$, is positive. For the natural-gas calendar, gross $+0.0004$ against cost 0.003 yields net -0.0026 : uneconomic.

Selection-on-deviation. Conditioning on $|z| \geq \theta$ samples directionally-efficient paths by construction; it is the complete and sufficient null for the killed pre-reversion morphology.

References

- [1] Working, H. (1949). The theory of price of storage. *American Economic Review*, 39(6), 1254–1262.
- [2] Szymanowska, M., de Roon, F., Nijman, T., & van den Goorbergh, R. (2014). An anatomy of commodity futures risk premia. *Journal of Finance*, 69(1), 453–482.
- [3] Lo, A. W., & MacKinlay, A. C. (1988). Stock market prices do not follow random walks: evidence from a simple specification test. *Review of Financial Studies*, 1(1), 41–66.

- [4] Lo, A. W., & MacKinlay, A. C. (1990). When are contrarian profits due to stock market overreaction? *Review of Financial Studies*, 3(2), 175–205.
- [5] George, T. J., & Hwang, C.-Y. (2004). The 52-week high and momentum investing. *Journal of Finance*, 59(5), 2145–2176.
- [6] Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- [7] Engle, R. F., & Granger, C. W. J. (1987). Co-integration and error correction: representation, estimation, and testing. *Econometrica*, 55(2), 251–276.
- [8] McLean, R. D., & Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1), 5–32.